

	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE	
	Supplimentary End Semester Examination – Summer 2022	
	Course: B. Tech.	Branch:
		Semester: II
	Subject Code & Name: BTES205E/BTE1205 Energy and Environment Engineering	
	Max Marks: 60	Date:
		Duration: 3 Hr.
	Instructions to the Students: 1. All the questions are compulsory. 2. Each question carries 12 marks. 3. Illustrate your answers with neat sketches, diagram etc., wherever necessary 4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 5. Use of non-programmable scientific calculators is allowed. 6. Assume suitable data wherever necessary and mention it clearly.	
		(Level/CO)
		Marks
Q. 1	Solve Any Two of the following.	
A)	What are the fossil fuels used for generation of conventional power? Compare the Gas Turbine Power Plant with the Coal based power plant in respect of site requirement, environmental effect, initial cost, fuel transportation, reliability, operating cost, simplicity, and cleanliness.	Remember/CO1 Understand/CO1
		2 4
B)	Explain the function of following components used in nuclear power plant: i) Moderator ii) Condenser iii) Reactor.	Understand/CO1
		6
C)	Show the classification of Hydroelectric power plants on the basis of availability of water head, water flow and loading type using suitable flow chart. Name the various elements of the Hydroelectric power plant and explain the following: Draft tube and Tail race.	Remember/CO1 Remember/CO1
		2 4
Q.2	Solve Any Two of the following.	
A)	What is solar energy? Define solar cell? Explain its working with a simple sketch.	Remember/CO2 Understand/CO2
		2 4
B)	What are the various sources of biomass? Explain with a simple diagram the working of a Bio gas plant in villages.	Remember/CO2 Understand/CO2
		2 4
C)	Write any four parameters that are to be considered while selecting site for wind mill installation. Differentiate between horizontal axis wind turbine and vertical axis wind turbine.	Understand/CO2 Understand/CO2
		2 4
Q. 3	Solve the following.	
A)	What do you mean by star rating of the home appliances? How it is decided? Discuss the energy conservation measures to be adopted for the Refrigerator at home.	Remember/CO3 Understand/CO3
		3 3
B)	Define energy efficiency. Write the various practices of making the home energy efficient in today's scenario.	Remember/CO3 Understand/CO3
		2 4

Q.4	Solve Any Two of the following.		
A)	Define air pollution. How the secondary air pollutants are produced? Give one example. What are their effects on human health?	Remember/CO4 Understand/CO4	2 4
B)	Name at least six indoor air pollutants. Explain the measures to be taken to reduce or prevent these pollutants from the home.	Remember/CO4 Understand/CO4	2 4
C)	What is Smog, and how is it created? Explain the effect of photochemical smog on the human health and also write the remedy of preventing the formation of this smog.	Remember/CO4 Understand/CO4	2 4
Q. 5	Solve Any Four of the following.		
A)	What are the types of noise pollution? How do you minimize the noise pollution from transportations?	Understand/CO5	3
B)	Give any three examples of monuments in India that are affected by the pollution.	Remember/CO5	3
C)	What is eutrophication and how can it be controlled?	Understand/CO5	3
D)	If you have observed any water pollution in your area then suggest how can you control it?	Understand/CO5	3
E)	What is the effect of noise on students? How it can be minimized?	Understand/CO5	3
F)	What are the sources of radioactive pollution? Why it is most harmful?	Remember/CO5	3
	*** End ***		

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